Examiner only

SECTION A

*Answer **all** questions in the spaces provided.*

1. When aqueous iodide ions are added to an aqueous solution of lead(II) nitrate, a precipitate is formed.

(a) Give the colour of the precipitate. [1]

....

(b) Write an **ionic** equation for the reaction that occurs, including state symbols. [1]

....

2. Give an example of a transition metal used as a catalyst. You should name the metal and identify the reaction it catalyses. [1]

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3. Calcium chloride, $CaCl_2$, is soluble in water. Use the values below to explain why this is the case. [2]

Equation 1 $Ca^{2+}(g) + 2Cl^{-}(g) \longrightarrow CaCl_2(s)$ $\Delta H^{\theta} = -2237\ kJ\ mol^{-1}$

Equation 2 $Ca^{2+}(g) + 2Cl^{-}(g) + aq \longrightarrow CaCl_2(aq)$ $\Delta H^{\theta} = -2378\ kJ\ mol^{-1}$

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(c) Nitrogen dioxide is used in the production of nitric acid. It is produced in two stages.

Stage 1 $4NH_3(g) + 5O_2(g) \rightleftharpoons 4NO(g) + 6H_2O(g)$ $\Delta H^\theta = -900\ kJ\ mol^{-1}$

Stage 2 $2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$ $\Delta H^\theta = -115\ kJ\ mol^{-1}$

(i) Explain why the use of a catalyst is essential in industrial processes involving exothermic equilibria such as stage 1. [2]

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(ii) A student attempted to perform stage 2 using a temperature of 450 K and a pressure of 5 atm. He obtained only a small yield. Suggest how the yield could be improved, explaining your reasons. [4]

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SECTION A

Answer **all** *questions in the spaces provided.*

1. Give the electronic configuration of a copper atom. [1]

....

2. Excess sodium hydroxide is added to an aqueous solution of chromium(III) chloride, $CrCl_3$. Give the formula of the chromium-containing ion formed. [1]

....

3. *(a)* Write the equation for the reaction of hot concentrated aqueous sodium hydroxide with chlorine. [1]

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(b) This is an example of a disproportionation reaction. State what is meant by the term *disproportionation*. [1]

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4. Give the observation(s) expected when water is added to $SiCl_4$. [1]

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SECTION A

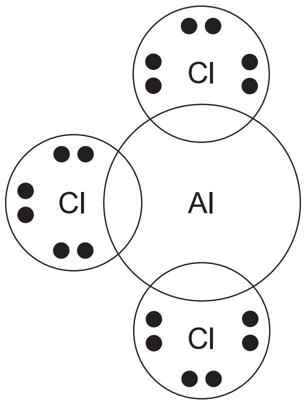
*Answer **all** questions in the spaces provided.*

1. Complete the electronic structure of the Co^{2+} ion below. [1]

$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$ ....

2. The compound $AlCl_3.NH_3$ contains both covalent and coordinate bonds.

Complete the dot and cross diagram for this compound. [2]



3. Write the expression for the ionic product of water, K_w. [1]

4. The standard hydrogen electrode and hydrogen fuel cells both use the same metal as an electrode. Name the metal and explain why this metal is suitable. [1]

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Examiner only

SECTION A

Answer **all** questions.

1. Addition of aqueous lead(II) nitrate to aqueous potassium iodide causes a precipitate to form.

(a) Give the colour of the precipitate. [1]

....

(b) Write an ionic equation for the reaction. [1]

....

2. Addition of excess hydrochloric acid to aqueous copper(II) sulfate causes the solution to turn green.

Give the formula of the copper-containing species present. [1]

....

3. The rate equation for the decomposition of N_2O_5 to form O_2 and NO_2 is first order overall.

(a) Write the rate equation for this reaction. [1]

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(b) Suggest a rate-determining step for this reaction. [1]

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4. State what is meant by the term 'standard electrode potential'. Include the conditions required. [2]

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Examiner only

SECTION A

Answer **all** questions.

1. Addition of aqueous lead nitrate to aqueous potassium iodide causes a precipitate to form. Write an ionic equation for the formation of the precipitate, including state symbols. [1]

....

2. Use the data shown to explain why silver chloride is insoluble in water. [2]

	Enthalpy / kJ mol^{-1}
Standard enthalpy of hydration for Ag^+	−464
Standard enthalpy of hydration for Cl^-	−364
Enthalpy of lattice formation of AgCl	−905

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3. A white crystalline solid gives a red colour in a flame test and addition of concentrated sulfuric acid to the solid gives purple fumes and a smell of rotten eggs.
Name the ions present, giving your reasons. [2]

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4. Aluminium chloride forms Al_2Cl_6 dimers using coordinate bonding. Explain why the coordinate bonds form. [2]

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